

Evaluating Dataset Scaling and Edge Case Vulnerabilities in Neural Network-Based ITSM Ticket Classification

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Abstract— In enterprise Information Technology Service Management (ITSM) operations, streamlining incident triage and end-user feedback processing demands flexible text classification architectures capable of moving beyond rigid keyword rules. This study evaluates a neural network-based text classifier engineered to dynamically categorize operational feedback into three distinct classes: Positive Praise, Negative Complaint, and Urgent Support. The system was tested iteratively to measure predictive confidence against standard incident reports, sarcastic phrasing, and out-of-scope administrative inquiries. The empirical findings demonstrate both the efficacy and limitations of the architecture: scaling the training dataset from 30 to 180 examples substantially improved classification reliability, increasing average confidence for standard inputs (from 70.2% to 85.0%) and successfully correcting sarcastic anomalies (from 61.0% to 80.2%). However, testing also uncovered a critical vulnerability regarding out-of-vocabulary inputs, where unrepresented administrative edge cases were mathematically coerced into existing operational parameters with an average false-positive confidence of 96.6%. These results confirm that deploying reliable neural networks in dynamic ITSM environments requires a combined approach of continuous, representative dataset scaling and explicit programmatic safeguards for unclassified inputs.

Keywords—Automated Triage, Information Technology Service Management (ITSM), Machine Learning, Natural Language Processing (NLP), Neural Networks, Text Classification.

I. INTRODUCTION

In modern enterprise Information Technology Service Management (ITSM) and IT Outsourcing (ITO) environments, service desks process continuous streams of end-user feedback and incident tickets aligned with standard ITIL frameworks [1]. Educational platforms, such as Machine Learning for Kids, introduce foundational artificial intelligence concepts by providing a no-code web-based environment where text recognition models can be constructed without extensive programming [2]. While traditional semantic analysis relies heavily on hand-coded keywords and hardcoded rule-based logic, modern neural networks construct text classifiers dynamically by converting text into numerical tokens and computing mathematical associations between word patterns and predefined labels [3].

Despite the operational advantages of neural network-based Natural Language Processing, classification accuracy remains highly dependent on the volume, quality, and representative diversity of the dataset [4]. High computational complexity arises when handling out-of-

vocabulary terms, sarcastic phrasing, or ambiguous end-user complaints [5]. When encountering text patterns that lack contextual representation in the training set, the neural network often yields split confidence scores across multiple categories, leading to the misclassification of critical IT support tickets [6]. Therefore, evaluating how variations in training data volume and composition influence model confidence is essential for ensuring reliable automated triage [7].

To address these limitations, this paper details the implementation and evaluation of a neural network-based text classifier using the Machine Learning for Kids platform to process ITSM end-user feedback. The model categorizes incoming technical feedback into three distinct operational classes: Positive Praise, Negative Complaint, and Urgent Support. The primary contributions of this work include demonstrating the direct correlation between dataset size and model confidence, identifying system failure modes through stress-testing against ambiguous inputs, and executing a model refinement iteration to correct observed misclassifications and optimize predictive performance [8].

II. SYSTEM DESIGN AND DATA COLLECTION

A. Architectural Pipeline and Storage

The modeling dashboard utilized for this experiment is structured into three primary operational stages. The first stage, designated as the Train phase, serves as the data collection module where tagged examples are defined and grouped into specific intent categories. Following this, the Learn and Test phase executes the core training process, allowing the neural network to update its internal weights based on the ingested dataset. Finally, the Make phase operates as the inference engine, where live, unseen text inputs are evaluated against the trained model to test classification accuracy. For the duration of this setup, the project configuration specifies the local web browser as the primary storage mechanism.

B. Dataset Construction and Categorization

To establish a functional baseline model, the training dataset was populated with a minimum of ten distinct text examples for each of the three predefined operational categories. The Urgent Support dataset was constructed using critical incident scenarios, focusing on immediate operational blockers such as enterprise system outages, authentication failures, and critical hardware disconnections. The Negative Complaint dataset incorporated reports of system friction, highlighting end-user issues like software

interface inefficiencies, unexpected application restarts, and persistent network latency. Conversely, the Positive Praise dataset featured successful operational deployments, containing feedback regarding rapid incident resolution, seamless infrastructure migrations, and highly effective technical support.

C. Model Training Execution

During the Learn and Test phase, the platform dynamically compiles the text sentiment classifier by processing the categorized datasets. The system executes this by converting the provided textual examples into numerical tokens and subsequently mapping the mathematical associations between these word patterns and the designated class labels. This execution phase ensures that the model can perform automated feature extraction purely based on the tagged training data, validating the architecture's capacity to execute without relying on explicit, manually coded logic.

III. TESTING AND EVALUATION

A. Baseline Inference Testing

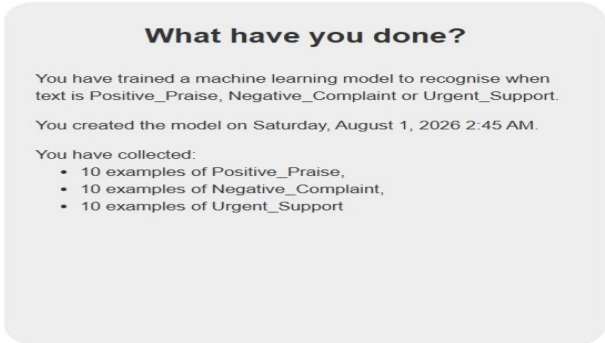


Fig. 1. Baseline model training dataset summary.

In the initial testing phase shown in Fig.1, the baseline model was trained using a minimal dataset of 30 total examples, distributed equally with ten items across each of the three operational categories. Testing with clear pattern phrases demonstrated functional but inconsistent classification capabilities. While the model correctly categorized standard incidents and praises, the predictive confidence scores remained predominantly low, generally ranging between 52% and 76%. Furthermore, the system experienced unexpected misclassifications during this baseline phase, such as incorrectly identifying a positive review regarding a new self-service catalog as a Negative Complaint with a low confidence score of 63%.

B. Stress Testing and Data Scaling Iterations

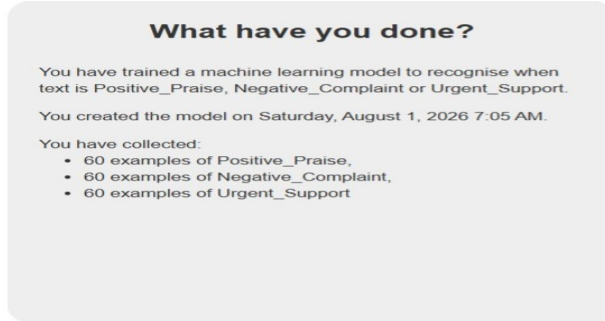


Fig. 2. Scaled model training dataset summary.

When subjected to ambiguous and sarcastic phrasing during the baseline iteration, the model's accuracy degraded significantly. Phrases containing contradictory sentiment markers, such as praising a software patch that caused a system outage, were incorrectly classified as Positive Praise due to the presence of positive vocabulary. To address this failure mode, a subsequent iteration expanded the training dataset significantly to 180 examples, providing 60 items per category, as shown in Fig. 2. This deliberate data scaling successfully corrected the sarcastic phrasing failures; the retrained model correctly identified all sarcastic inputs as Negative Complaints, with peak confidence scores improving to 97%. This demonstrates how scaling data volume and representative diversity directly improves mathematical mapping and predictive accuracy.

C. Edge Case and Out-of-Vocabulary Evaluation

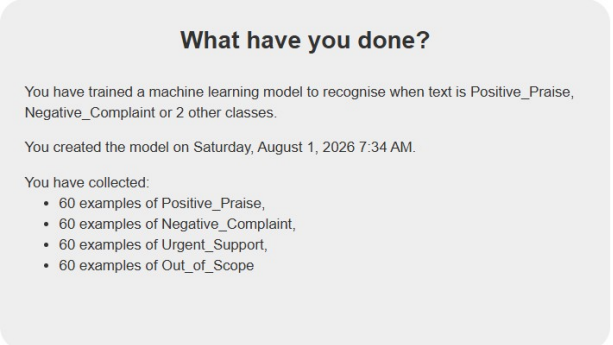


Fig. 3. Refined model training dataset summary.

To further evaluate the system's robustness, a final testing iteration introduced out-of-scope administrative inquiries, utilizing questions regarding cafeteria payments, transit subsidies, and room reservations that lacked specific contextual representation in the training data. The neural network forcefully mapped these unclassified administrative requests to the Negative Complaint category with abnormally high confidence scores ranging from 93% to 99%. This observation highlights a critical failure mode: when a neural network lacks contextual representation for general inquiries, it mathematically coerces the out-of-vocabulary input into the closest existing class. This confirms that without specific training data for edge cases, the model is prone to high-confidence misclassifications, necessitating continuous evaluation and dataset refinement.

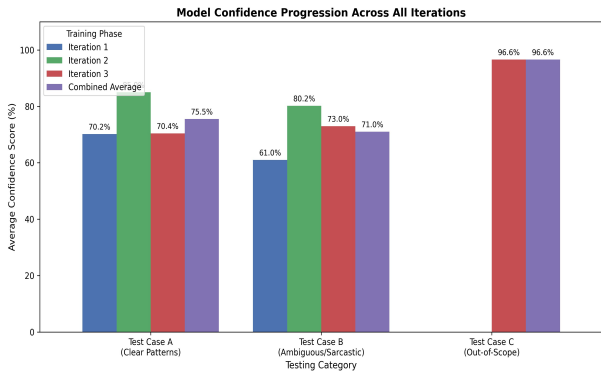


Fig. 4. Model confidence progression across testing iterations.

As illustrated in Fig. 4, the progression of average confidence scores reveals both the benefits of dataset scaling and a critical vulnerability within the neural network classifier. While scaling the training volume between Iteration 1 and Iteration 2 substantially improved predictive reliability for standard classification (from 70.2% to 85.0%) and ambiguous phrasing (from 61.0% to 80.2%), the introduction of unrepresented administrative edge cases in Iteration 3 induced a slight regression in these established categories. Most notably, the model exhibited a severe operational vulnerability when processing out-of-vocabulary inputs (Test Case C); rather than failing to classify unknown variables, the neural network forcefully mapped these out-of-scope inquiries into existing operational parameters with an overwhelmingly high false confidence of 96.6%. The combined averages ultimately stabilize at 75.5% and 71.0% for recognized text patterns, yet the model's near-perfect false confidence for untrained data emphasizes the critical need for explicit programmatic safeguards or a designated "unclassified" category in dynamic ITSM environments.

IV. DISCUSSION AND ANALYSIS

A. Explicit Rules Versus Tagged Learning

Relying on explicit conditional logic to categorize end-user complaints presents significant scalability and maintenance challenges within enterprise IT operations. If an ITSM platform attempted to process all possible incident reports using hardcoded logic, the system would require an impractical volume of manual rule configurations to account for typographical errors, regional dialects, and evolving technical terminology. In contrast, tagged learning allows the neural network to dynamically map relationships between words without requiring predefined dictionaries. This provides a highly flexible and sustainable architecture capable of processing diverse natural language inputs without continuous manual intervention.

B. Data Bias and Dataset Quality

The accuracy of deep learning models relies heavily on the quality, volume, and representative diversity of the training data. Operating with an insufficient dataset prevents the neural network from establishing reliable mathematical associations, resulting in low predictive confidence and frequent misclassifications. Expanding the dataset with diverse contextual examples provides the model with the necessary variations to recognize nuanced phrasing and mitigate inherent data bias. As observed during testing, high data volume and diverse representation are critical for the model to successfully categorize complex edge cases, such as sarcastic technical complaints, that would otherwise trick a limited baseline model.

C. Feature Extraction Pipeline Architecture

Within the constructed modeling pipeline, feature extraction occurs explicitly during the Learn and Test phase. As the platform ingests the tagged training examples, the neural network initiates the process by converting the raw textual data into numerical tokens. It is during this tokenization and the subsequent adjustment of internal weights that the system autonomously identifies the defining characteristics of each intent category. By calculating these mathematical associations internally, the model successfully

extracts the necessary linguistic features to perform automated sentiment classification during the final inference phase.

V. CONCLUSION & FUTURE WORK

This study demonstrated the implementation and evaluation of a neural network designed to classify ITSM end-user feedback into distinct operational categories without relying on hardcoded rule-based logic. By utilizing a dynamic learning architecture, the model successfully extracted features and established mathematical associations purely from tagged text examples. The experimental results confirmed that the predictive accuracy and confidence of deep learning models are directly dependent on the volume, quality, and representative diversity of the training dataset.

While the baseline model struggled to categorize ambiguous and sarcastic phrasing correctly, intentionally scaling the data volume resolved these failure modes and significantly improved classification reliability. Furthermore, edge case testing revealed a critical vulnerability where the neural network forcefully mapped out-of-vocabulary administrative inquiries into existing categories with abnormally high confidence. This confirms that when a model lacks contextual representation for general inquiries, it is highly susceptible to misclassification errors.

Ultimately, deploying automated Natural Language Processing for enterprise service desk operations requires continuous evaluation and dataset refinement. To maintain reliable automated triage, organizations must ensure their training datasets comprehensively represent the diverse and evolving natural language patterns present in daily end-user feedback.

REFERENCES

- [1] Alsac, A., Yilmaz, U., Koçoglu, F. Ö., et al. (2025). Towards Intelligent IT Service Management: A Comparative Evaluation of Machine Learning and Language Models for Ticket Classification and Assignment. IEEE. DOI: 10.1109/ubmk67458.2025.11207046
- [2] Ouyang, F., Jiao, L., & Luo, M. (2022). Artificial intelligence (AI) learning tools in K-12 education: A scoping review. *Interactive Learning Environments*. DOI: 10.1080/10494820.2022.2043908
- [3] Ahmed, S., Singh, M., Doherty, B., et al. (2023). Knowledge-based Intelligent System for IT Incident DevOps. 2023 IEEE International Conference on AIOps. DOI: 10.1109/aio59134.2023.00005
- [4] Koukal, F., Dařena, F., Jeřdík, R., et al. (2024). Improving Automated Categorization of Customer Requests with Recent Advances in Natural Language Processing. *European Journal of Business Science and Technology*. DOI: 10.11118/ejobsat.2024.010.
- [5] Liu, Z. (2023). Ticket-BERT: Labeling Incident Management Tickets with Language Models. arXiv preprint. DOI: 10.48550/arXiv.2307.00108.
- [6] (2024). A Contextualized Deep Neural Network Model for Classifying IT Incidents Based on Severity. 2024 IEEE International Conference on Computing and Artificial Intelligence (ICCA). DOI: 10.1109/icca62237.2024.10927882.
- [7] Paket, E., Şenerkek, G., Akyol, B., et al. (2024). IT Service Desk Ticket Classification via Large Language Models. 2024 6th International Conference on Computer Science and Engineering (UBMK). DOI: 10.1109/ubmk63289.2024.10773473.
- [8] Abd, S., Al-Attar, B., Yousif, H. A., et al. (2025). Automated Multimodal Ticket Classification in Public ITSM Systems Using Hierarchical Attention and Deep Neural Networks on Mixed-Format Inputs. 2025 International Conference on Computing and Robotics (ICCR). DOI: 10.1109/iccr67387.2025.11292007